

1. Introduction to C# and .NET 8

What is C#?

C# (C-Sharp) is a modern **object-oriented programming language** developed by Microsoft.

It is used to build:

- Desktop Applications (WPF, WinForms)
- Web Applications (ASP.NET Core)
- APIs and Microservices
- Cloud Applications (Azure)
- Mobile Apps (.NET MAUI)
- Game Development (Unity)

Example companies using C#:

- Microsoft
- StackOverflow
- Accenture
- Many enterprise banking systems

What is .NET?

.NET is a **software development platform**.

It provides:

1. **Runtime** – executes programs
 2. **Libraries** – ready-made functionality
 3. **Tools** – compilers, SDKs, CLI
 4. **Frameworks** – ASP.NET, MAUI, WPF
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.NET 8 Architecture

When you write a C# program, this is what happens internally.

C# Code



C# Compiler (Roslyn)



Intermediate Language (IL)



CLR (Common Language Runtime)



JIT Compilation



Machine Code



Execution on OS

Key Components

Component	Purpose
CLR	Executes .NET programs
BCL	Base Class Library
JIT	Converts IL to machine code
GC	Garbage Collection (memory management)

.NET 8 Project Structure

When you create a **Console App in Visual Studio**:

File → New → Project → Console App (.NET)

You will see a structure like this:

```
MyFirstApp
|
├── Program.cs
├── MyFirstApp.csproj
├── bin/
└── obj/
```

Program.cs (Entry Point)

In .NET 8, the program can start without the `Main` method.

Example:

```
Console.WriteLine("Hello .NET 8!");
```

Visual Studio automatically generates the entry point.

2. C# Syntax and Basic Constructs

Example Full Program

Create a **Console App (.NET 8)**

Replace Program.cs with this:

```
using System;

class Program
{
    static void Main()
    {
        Console.WriteLine("Welcome to C# Programming!");
    }
}
```

```
int age = 25;
string name = "Ganesh";
double salary = 45000.50;

Console.WriteLine("Name: " + name);
Console.WriteLine("Age: " + age);
Console.WriteLine("Salary: " + salary);
}
}
```

Output

```
Welcome to C# Programming!
Name: Ganesh
Age: 25
Salary: 45000.5
```

3. Data Types in C#

Data types define **what type of data a variable can store**.

Value Types

Stored directly in memory.

Type	Description	Example
int	Integer	10
double	Decimal number	10.5
bool	True/False	true
char	Single character	'A'

Example:

```
int age = 30;
double price = 99.99;
char grade = 'A';
bool isActive = true;
```

Reference Types

Stored as **memory reference**.

Examples:

- string
- class
- object

Example:

```
string name = "Ganesh";
```

4. Variables

A **variable** is a named memory location used to store data.

Syntax

```
datatype variableName = value;
```

Example

```
int marks = 95;  
string city = "Chennai";  
double temperature = 32.5;
```

Example Program

using System;

```
class Program  
{  
    static void Main()  
    {  
        int id = 101;  
        string name = "Ravi";  
        double salary = 55000;
```

```
    Console.WriteLine("Employee ID: " + id);  
    Console.WriteLine("Employee Name: " + name);  
    Console.WriteLine("Employee Salary: " + salary);  
}  
}
```

5. Control Flow Statements

Control flow statements **control the order in which code executes**.

Types:

1. if-else
 2. switch
 3. loops
 4. conditional operators
-

6. IF-ELSE Statement

Used to execute code based on a **condition**.

Syntax

```
if(condition)  
{  
    code  
}  
else  
{  
    code  
}
```

Example 1 – Age Check

using System;

```
class Program
{
    static void Main()
    {
        int age = 20;

        if (age >= 18)
        {
            Console.WriteLine("You are eligible to vote.");
        }
        else
        {
            Console.WriteLine("You are not eligible to vote.");
        }
    }
}
```

Output

You are eligible to vote.

Example 2 – Multiple Conditions

using System;

```
class Program
{
    static void Main()
    {
        int marks = 75;

        if (marks >= 90)
        {
            Console.WriteLine("Grade A");
        }
        else if (marks >= 75)
        {
            Console.WriteLine("Grade B");
        }
        else if (marks >= 50)
```

```
{  
    Console.WriteLine("Grade C");  
}  
else  
{  
    Console.WriteLine("Fail");  
}  
}  
}
```

7. SWITCH Statement

Used when there are **multiple fixed conditions**.

Example: menu systems.

Syntax

```
switch(variable)  
{  
    case value:  
        code  
        break;  
}
```

Example – Day of Week

using System;

```
class Program  
{  
    static void Main()  
    {  
        int day = 3;  
  
        switch (day)  
        {  
            case 1:  
                Console.WriteLine("Monday");  
                break;  

```

```
        case 2:
            Console.WriteLine("Tuesday");
            break;

        case 3:
            Console.WriteLine("Wednesday");
            break;

        default:
            Console.WriteLine("Invalid Day");
            break;
    }
}
```

Output

Wednesday

8. Loops

Loops repeat a block of code.

Types:

1. for loop
 2. while loop
 3. do-while loop
 4. foreach loop
-

9. FOR LOOP

Used when **number of iterations is known**.

Syntax

for(initialization; condition; increment)

Example – Print Numbers

using System;

class Program

```
{
    static void Main()
    {
        for(int i = 1; i <= 5; i++)
        {
            Console.WriteLine("Number: " + i);
        }
    }
}
```

Output

Number: 1
Number: 2
Number: 3
Number: 4
Number: 5

10. WHILE LOOP

Runs **while** condition is true.

Example

using System;

class Program

```
{
    static void Main()
    {
        int i = 1;

        while(i <= 5)
        {
            Console.WriteLine("Count: " + i);
        }
    }
}
```

```
        i++;  
    }  
}  
}
```

11. DO-WHILE LOOP

Executes **at least once**.

Example

using System;

```
class Program  
{  
    static void Main()  
    {  
        int i = 1;  
  
        do  
        {  
            Console.WriteLine("Value: " + i);  
            i++;  
        }  
        while (i <= 5);  
    }  
}
```

12. FOREACH LOOP

Used to iterate collections.

Example

using System;

```
class Program  
{  
    static void Main()  
    {
```

```
string[] fruits = { "Apple", "Mango", "Orange" };

foreach(string fruit in fruits)
{
    Console.WriteLine(fruit);
}
}
```

Output

Apple
Mango
Orange

13. Conditional Operator (Ternary Operator)

Short form of **if-else**.

Syntax

condition ? trueValue : falseValue

Example

```
using System;

class Program
{
    static void Main()
    {
        int age = 16;

        string result = (age >= 18) ? "Adult" : "Minor";

        Console.WriteLine(result);
    }
}
```

Output

14. Real-World Example (Ready to Run)

Example: **Simple Student Result Program**

using System;

```
class Program
{
    static void Main()
    {
        Console.Write("Enter Student Name: ");
        string name = Console.ReadLine();

        Console.Write("Enter Marks: ");
        int marks = Convert.ToInt32(Console.ReadLine());

        string grade;

        if (marks >= 90)
            grade = "A";
        else if (marks >= 75)
            grade = "B";
        else if (marks >= 50)
            grade = "C";
        else
            grade = "Fail";

        Console.WriteLine("Student: " + name);
        Console.WriteLine("Marks: " + marks);
        Console.WriteLine("Grade: " + grade);
    }
}
```

Summary

You learned:

.NET Platform

- .NET Runtime
- CLR
- JIT
- Libraries

C# Basics

- Data types
- Variables
- Syntax
- Program structure

Control Flow

- if-else
 - switch
 - loops
 - conditional operator
-